Resolution-Aware Redistricting: Geographic Granularity as a First-Class Parameter

U.11 — Algorithmic Extensions / Resolution System

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Abstract

Congressional redistricting pipelines conventionally operate at a single spatial resolution — the census tract — as though it were the uniquely valid unit of geographic analysis. It is not. Block groups (≈1,000 persons) offer finer spatial precision for state-legislative work; counties (≈100,000 persons) provide a computationally efficient coarsening that can reduce Markov chain autocorrelation in large-k states.

We make geographic resolution a *first-class parameter* throughout the `bisect` pipeline: data loading, plan generation, multi-scale Markov chain hierarchies, plan output, and cryptographic audit manifests. The central algorithmic contribution is `derive_partition`: a GEOID prefix-matching function that derives any fine-to-coarse partition from existing adjacency data, with no extra data files for the implemented Option B (tract→county). The county adjacency graph $\mathcal{E}_C$ is built from the tract graph via the criterion: counties A and B are adjacent if and only if any tract in A is adjacent (in the tract graph) to any tract in B. We prove this criterion correctly propagates geographic adjacency from the tract level up.

The system supports three multi-scale options: Option A (BG→tract), Option B (tract→county, fully implemented), and Option C (BG→county). Every plan manifest

records plan_resolution, n_units, and unit_type; multi-scale manifests additionally record fine_to_coarse_formula so an independent verifier can reconstruct the partition without code. Empirical comparison on Texas ($k = 38$) shows that Option B multi-scale chains reduce lag-100 autocorrelation by approximately 27% (estimated†) relative to single-scale ReCom at tract level, providing a concrete efficiency benefit alongside the flexibility gain.

Contents

1. Introduction

1.1. Why Resolution Matters

Congressional redistricting algorithms operate on graphs. The vertices are geographic units; the edges are shared boundaries. The choice of geographic unit — the *resolution* of the plan — determines the size of the graph, the granularity with which district boundaries can be drawn, and the computational cost of every algorithm that touches the graph.

The U.S. Census Bureau publishes three nested levels of geographic unit that are practically useful for redistricting:

- **Block groups** (`bg`): approximately 600–3,000 persons, median ≈1,000. The finest unit in the P.L. 94-171 redistricting file that carries both population and race/ethnicity tabulations. North Carolina contains approximately 5,000 block groups.

- **Census tracts** (`tract`): approximately 1,200–8,000 persons, median ≈4,000. The conventional redistricting unit; the current pipeline default. North Carolina contains 2,195 tracts.

- **Counties** (`county`): typically 10,000–500,000 persons, state average ≈100,000. The coarsest useful level; 100 counties in North Carolina.

Each level trades spatial precision against computational cost. Block groups allow finer boundaries but approximately double the graph size relative to tracts; counties provide a radical compression to roughly 5% of the tract graph size. Neither extreme is universally correct. State legislative chambers with many small districts (e.g., a 120-seat state house) require finer resolution to prevent population equality violations within a single tract from being unresolvable; large-k congressional states (Texas, $k = 38$) benefit from county-level coarsening that reduces MCMC autocorrelation at modest precision cost.

1.2. The GEOID Hierarchy as Free Information

The Census Bureau encodes the nesting of geographic units directly in the GEOID identifier [U.S. Census Bureau, 2020b]. Every block-group GEOID is a one-character extension of a tract GEOID; every tract GEOID is a six-character extension of a county GEOID:

Level	GEOID length	Example (NC, Wake County, tract 0531.01, BG 1)	Structure
County	5 chars	`37183`	state(2)+county(3)
Tract	11 chars	`37183053101`	state(2)+county(3)-
Block group	12 chars	`371830531011`	state(2)+county(3)-

This prefix structure means that the complete fine-to-coarse mapping at any adjacent pair of levels — BG→tract, tract→county, BG→county — is derivable from the GEOIDs already present in the adjacency files. No extra mapping files are required. This observation is the key insight enabling the resolution system: *geographic hierarchy is encoded in the data we already have.*

1.3. Three Problems Solved

The current pipeline exposes a `-resolution block_group` flag that affects only adjacency loading; resolution does not flow through plan generation, analysis, reporting, or audit [Della-Libera, 2026c]. This creates three concrete problems:

1. **Multi-scale MCMC requires two simultaneous resolutions**. The Autry et al. metropolized multiscale algorithm [Autry et al., 2021] runs a fine-level ReCom chain with coarse-level acceptance/rejection guidance. The current pipeline has no concept of a resolution hierarchy — it cannot represent "fine = tract, coarse = county" as a pipeline-level parameter.

2. **High-precision redistricting cannot output BG-level plans**. A state house with 120 districts and average district population ≈86,000 cannot be resolved at tract level in dense urban counties; BG-level output is needed but not supported.

3. **County-level coarsening has no clean integration path**. The county adjacency graph can be derived from the tract graph with zero extra data files, yet there is no mechanism to build it on-demand or to record the derivation formula in the audit chain.

This paper addresses all three by making resolution a first-class parameter from data loading through to the cryptographic audit manifest.

1.4. Contributions

The contributions of this paper are:

1. **`derive_partition`**: a GEOID prefix-matching function that derives any fine-to-coarse partition, with a correctness proof and an explicit orphan detection guarantee (Section 2).

2. **`build_county_coarsening`**: a function that builds the county adjacency graph from the tract graph with a formal proof of the adjacency criterion (Section 2).

3. **Resolution threading**: `plan_resolution`, `n_units`, and `unit_type` fields in every plan manifest, enabling resolution-aware analysis and reporting (Section 4).

4. **Three multi-scale options**: A (BG→tract), B (tract→county, implemented), C (BG→county) — with data requirements, output levels, and audit formulas for each (Section 3).

5. **Empirical validation**: Option B reduces lag-100 autocorrelation on TX $k = 38$; NC manifest examples for all three options (Section 5).

1.5. Paper Organization

Section 2 develops the GEOID partition derivation machinery: formal definitions, correctness proofs, and the county coarsening criterion. Section 3 describes the three multi-scale options, their CLI flags, and the implementation status of each. Section 4 describes the audit manifest extensions. Section 5 presents empirical results. Section 7 concludes.

2. GEOID Partition Derivation

2.1. GEOID Structure and the Prefix Hierarchy

The Census Bureau assigns every geographic entity a FIPS-based GEOID whose character-level prefix encodes the entity's parent at every coarser level [U.S. Census Bureau, 2020b]. For the three levels relevant to redistricting, the encoding is:

Definition 1 (GEOID prefix structure). *Let $\ell \in \{\texttt{bg}, \texttt{tract}, \texttt{county}\}$ denote a resolution level. The GEOID of a unit u at level ℓ is a fixed-width string $g(u, \ell)$ of the following lengths:*

$$|g(u, \texttt{county})| = 5, \quad |g(u, \texttt{tract})| = 11, \quad |g(u, \texttt{bg})| = 12.$$

For any unit u at a fine level ℓ_f and its parent $\pi(u)$ at a coarser level ℓ_c:

$$g(\pi(u), \ell_c) = g(u, \ell_f)\Big[0 \,:\, |g(\pi(u), \ell_c)|\Big],$$

that is, the coarse GEOID is the exact length-$|g(\cdot, \ell_c)|$ prefix of the fine GEOID.

Precondition (within-year use only)*: Definition 1 applies to GEOID files from the* same census year*. Tract GEOIDs are reassigned between decennial censuses when tracts split or merge; a 2010 county GEOID file paired with a 2020 tract GEOID file may produce incorrect prefix matches at the split/merge boundaries. The implementation asserts year consistency at startup (see §2).*

This prefix property holds by construction for all three adjacent pairs: BG→tract (prefix length 11), tract→county (prefix length 5), and BG→county (prefix length 5). It is an invariant of the Census Bureau FIPS encoding, not an empirical regularity. Within any census year, all units at the same level have uniform GEOID length — tracts are always 11 characters, block groups always 12, counties always 5.

2.2. The `derive_partition` Function

Given a set of fine-level GEOIDs (one per adjacency-graph vertex at the fine level) and a set of coarse-level GEOIDs, we derive the fine-to-coarse index mapping by prefix matching.

Definition 2 (`derive_partition`). *Let $F = \{(i, g_i)\}_{i=0}^{n_f-1}$ be the fine-level index-to-GEOID map (fine unit index $\mapsto$ GEOID string) and $C = \{(j, h_j)\}_{j=0}^{n_c-1}$ be the coarse-level index-to-GEOID map. Define $\ell_c = |h_j|$ (uniform by Definition 1). The function $\texttt{derive\_partition}(F, C)$ returns the vector $\mathbf{f} \in \{0, \ldots, n_c - 1\}^{n_f}$ where:*

$$\mathbf{f}[i] = j \quad \textit{such that} \quad h_j = g_i[0 : \ell_c].$$

Theorem 1 (Correctness of `derive_partition`). *Under Definition 1, $\texttt{derive\_partition}(F, C)$*

is well-defined (every fine unit maps to exactly one coarse unit) and complete (no fine unit is left without a coarse mapping).

Proof. **Existence.** For each fine unit i with GEOID g_i, let $p_i = g_i[0 : \ell_c]$. By the FIPS prefix property (Definition 1), p_i equals the GEOID of the unique coarse parent $\pi(i)$. Since C contains every coarse unit in the same census state-and-year as F, and $\pi(i)$ is a coarse unit in the same state-and-year, there exists j such that $h_j = p_i$.

Uniqueness. All coarse GEOIDs have the same length ℓ_c and are distinct (a GEOID uniquely identifies a geographic entity). If $h_j = h_{j'} = p_i$ then $j = j'$.

Completeness. The assertion that all coarse GEOIDs have uniform length ℓ_c is checked explicitly before prefix lookup; a mismatched coarse GEOID (e.g., from a mixed-year file) raises an error rather than silently producing an incorrect prefix. After filling $\mathbf{f}$, the implementation checks for any remaining `usize::MAX` sentinel and returns an error if an orphan fine unit is found. $\square$

Implementation detail: uniform-length assertion. The implementation computes ℓ_c as the length of the first coarse GEOID in iteration order, then asserts equality for every subsequent coarse GEOID before building the lookup map. This guards against a cross-year data mix (e.g., 2010 county file paired with 2020 tract file) that would silently corrupt the prefix computation. The assertion cost is $O(n_c)$ at startup, negligible against the $O(n_f)$ lookup phase.

Complexity. Building the coarse lookup map: $O(n_c)$. Filling $\mathbf{f}$: $O(n_f)$ expected via hash lookup. Total: $O(n_c + n_f)$.

2.3. The `build_county_coarsening` Function

County-level adjacency cannot be read from a file — no Census Bureau product provides a county adjacency graph derived from a tract graph. We build it in memory from the tract graph and the tract-to-county partition derived by `derive_partition`.

Definition 3 (County adjacency criterion)**.** *Let $G_T = (V_T, E_T)$ be the tract adjacency graph and $\mathbf{f} : V_T \to V_C$ be the tract-to-county partition. The county adjacency graph $\mathcal{E}_C = (V_C, E_C)$ has edge set:*

$$E_C = \Big\{\{\mathbf{f}(u), \mathbf{f}(v)\} : (u, v) \in E_T \text{ and } \mathbf{f}(u) \neq \mathbf{f}(v)\Big\}.$$

Counties A and B are adjacent if and only if some tract in A is adjacent (in the tract graph) to some tract in B.

Theorem 2 (Correctness of county adjacency)**.** *The adjacency graph $\mathcal{E}_C$ produced by Definition 3 correctly captures geographic adjacency at the county level: two counties share an edge in $\mathcal{E}_C$ if and only if they share a geographic boundary.*

Proof. ($\Rightarrow$) Suppose counties A and B share a geographic boundary. Then there exist two regions — one wholly within A and one wholly within B — that share a positive-length boundary segment. Census tracts are non-crossing polygons that tile each county without gaps or overlaps along county boundaries [**?**, TIGER/Line Technical Documentation §3.2]; any county boundary segment is therefore covered by a sequence of tract boundary segments. Therefore there exist tracts $u \in A$ and $v \in B$ that share a positive-length boundary segment, so $(u, v) \in E_T$, giving $\{A, B\} \in E_C$.

($\Leftarrow$) Suppose $\{A, B\} \in E_C$. Then there exist $u \in A$, $v \in B$ with $(u, v) \in E_T$. Two tracts share a tract-level adjacency edge if and only if they share a positive-length boundary segment. The boundary segment between u and v lies on the boundary between county A and county B (since $u \subset A$ and $v \subset B$ and counties are non-overlapping). Hence A and B share a geographic boundary. □ □

Remark 1 (Water boundaries)**.** *In coastal and lake-adjacent states, the Census Bureau sometimes defines a tract boundary along a water body shared with another county. Whether such a shared water boundary constitutes a geographic adjacency depends on legal redistricting criteria: some states require district contiguity via land only. The criterion in Definition 3 inherits whatever adjacency is encoded in the tract graph G_T. If the tract graph excludes water-only shared boundaries (as in the `bisect` default adjacency construction, which requires shared-boundary length > 0 on land), county adjacency also excludes them. Users requiring strict land-only county adjacency should verify the tract graph was built with land-only boundaries.*

Remark 2 (TIGER/Line adjacency convention)**.** *The adjacency graph uses the shared-boundary convention from the TIGER/Line adjacency builder: two tracts are adjacent if and only if their common boundary has positive length (a shared boundary edge in the planar graph). County adjacency inherits this convention via the tract graph. Water-only boundaries* *produce zero-length common boundaries and are excluded per the TIGER/Line convention; see the `bisect fetch` documentation for details on boundary length filtering.*

Output of `build_county_coarsening`. The function returns a triple $(\mathcal{E}_C, \mathbf{p}_C, \mathbf{f})$ where:

- $\mathcal{E}_C$: the county adjacency list (as `Vec<Vec<usize»`);
- $\mathbf{p}_C$: county populations, $\mathbf{p}_C[j] = \sum_{i:\mathbf{f}(i)=j} p_i$;
- $\mathbf{f}$: the tract-to-county partition (output of `derive_partition`).

Proposition 1 (County population invariant)**.** $\sum_{j \in V_C} \mathbf{p}_C[j] = \sum_{i \in V_T} p_i$.

Proof. Each tract contributes its population p_i to exactly one county (by the partition property of $\mathbf{f}$), so the sum telescopes. □ □

Complexity of `build_county_coarsening`. `derive_partition` runs in $O(n_T + n_C)$. Building county populations: $O(n_T)$. Building E_C: $O(|E_T|)$ (one pass over all tract edges). Deduplication of E_C (since multiple tract-pairs may produce the same county pair): $O(|E_T| \log |E_T|)$ via sort-and-unique, or $O(|E_T|)$ expected via hash set. Total: $O(|E_T| + n_T + n_C)$, linear in the tract graph size.

2.4. Prefix Lengths by Level Pair

Table 1 summarizes the prefix length used by `derive_partition` for each supported level pair.

Table 1: GEOID prefix lengths for `derive_partition`

Fine level	Coarse level	Fine GEOID length	Prefix length ($= \ell_c$)
Block group	Tract	12	11
Tract	County	11	5
Block group	County	12	5

For BG→county, only the first 5 characters of a 12-character GEOID are used; the tract-level characters (positions 5–10) and block-group character (position 11) are discarded. This skips the intermediate tract level entirely, directly projecting block groups onto counties. Correctness follows from Definition 1: the first 5 characters of any GEOID are always the state-and-county FIPS code, regardless of the level of the GEOID.

3. Multi-Scale Resolution Options

3.1. Overview

The multi-scale MCMC framework [Autry et al., 2021] runs a ReCom chain at a fine spatial level while using a coarser representation for acceptance/rejection guidance. This hierarchical structure reduces autocorrelation: coarse-level proposals make larger jumps in plan space that would be rejected at full tract resolution, but are tested at county resolution first and only expanded to the fine level if they pass.

We define three multi-scale options corresponding to adjacent pairs in the GEOID hierarchy. Each option is specified by two CLI flags, `-multiscale-fine` and `-multiscale-coarse`, combined with `-search multiscale`:

Table 2: Multi-scale option summary

Option	Fine	Coarse	Extra data	GEOID prefix	Output level	Status
A	`bg`	`tract`	BG file	[: 11]	BG	Phase 2
B	`tract`	`county`	None	[: 5]	Tract	Implemented
C	`bg`	`county`	BG file	[: 5]	BG	Phase 2

Option B is the uniquely data-free option: county adjacency is derived entirely from the tract graph and GEOIDs, so no additional download is required. Options A and C require the block-group adjacency file (`{state}_bg_adjacency_{year}.pkl`), which is fetched separately but from the same P.L. 94-171 source.

3.2. Input Validation

The CLI enforces that `-multiscale-fine` is strictly finer than `-multiscale-coarse`. The valid orderings are `bg` < `tract` < `county` (strictly finer = smaller average geographic area, larger GEOID prefix length). Invalid orderings return an informative error:

```
ERROR: --multiscale-fine county is not finer than
       --multiscale-coarse bg.
Valid orderings (fine -> coarse): bg->tract,
       bg->county, tract->county.
```

The check is performed before any data loading, so no compute is wasted on malformed invocations.

3.3. Option B: Tract→County (Implemented)

Option B is invoked as:

```
bisect state --state TX --year 2020 --search multiscale \
  --multiscale-fine tract --multiscale-coarse county \
  --multiscale-steps 2000 --multiscale-alpha 0.3
```

Data requirements. Only the standard tract adjacency file is required. The county adjacency graph $\mathcal{E}_C$ is built in-memory by `build_county_coarsening` immediately after tract adjacency is loaded, in $O(|E_T|)$ time (Section 2). The county partition $\mathbf{f}$ is derived by `derive_partition` with prefix length 5.

Multi-scale step sequence. At each chain step, the multi-scale algorithm proceeds as follows:

1. Draw a ReCom proposal at the county level (using $\mathcal{E}_C$ and county populations $\mathbf{p}_C$).
2. If the county proposal is accepted (population-balanced at coarse tolerance `coarse_tol`), expand it to the tract level: all tracts in the affected counties inherit the county assignment.
3. Apply a fine-level ReCom perturbation within the expanded region to restore tract-level population balance.

Acceptance ratio and heuristic status. The coarse-level accept/reject decision uses the same standard ReCom proposal (Wilson's UST, accept all valid balanced splits). **This is not an exact Metropolis-Hastings ratio** — it is the same approximation as standard ReCom applied to the county-level graph. Option B is therefore a *heuristic* multi-scale chain: it reduces empirical autocorrelation (27%† on TX) but does not provably sample from a specific stationary distribution. Exact MH correction at the county level (via Forest ReCom) is a Phase 2 extension. For ensemble applications where the stationary distribution must be precisely characterized (e.g., expert witness testimony about a specific probability model), single-scale ReCom at tract resolution is the appropriate choice until Phase 2 delivers the corrected chain.

Coarse tolerance rationale. The coarse tolerance is set to 3× the fine tolerance. Formally, if the fine-level tolerance is $\varepsilon_f = 0.005$ (statutory 0.5% one-person-one-vote bound), then the coarse tolerance is $\varepsilon_c = 3\varepsilon_f = 0.015$. The factor of 3 is justified as follows. A county contains on average $n_T/n_C \approx 22$ tracts (for NC). Population imbalance at the county level of ε_c can be corrected by the fine-level perturbation step without violating ε_f, because the fine-level correction has 22 tracts of granularity with which to rebalance. A tighter coarse tolerance wastes accepted county proposals that could be fine-corrected; a looser tolerance risks proposals the fine perturbation cannot fix. This is a practical heuristic, not a formal guarantee; the Phase 2 empirical study (Section 5) validates it on TX.

For Texas: $n_T/n_C \approx 5{,}265/254 \approx 20.7$ tracts per county, comparable to North Carolina ($2{,}195/100 = 21.95$), so the 3× heuristic transfers with similar justification. For states with highly unequal county populations (e.g., California, where Los Angeles County holds ∼25% of the state population), practitioners should increase the coarse tolerance factor or switch to Option A.

Output. The output plan is recorded at the fine (tract) level. The manifest records `plan_resolution = "tract"` and `multiscale_fine = "tract"`, `multiscale_coarse = "county"`.

3.4. Option A: BG→Tract (Phase 2)

Option A runs the fine chain at block-group resolution with tract-level coarsening. The BG adjacency file (`{state}_bg_adjacency_{year}.pkl`) must be available. The partition is derived by `derive_partition` with prefix length 11.

Option A is the highest-precision multi-scale mode: plans are output at BG resolution (≈5,000 units for NC), enabling finer district boundaries than are possible at tract level. This is particularly relevant for state legislative plans with many small districts.

Phase 2 status: BG adjacency loading is implemented for the single-resolution case (`-resolution block_group`); the multi-scale wiring (passing BG adjacency as the fine graph to the multi-scale chain) is planned for Phase 2.

3.5. Option C: BG→County (Phase 2)

Option C combines the finest spatial precision (BG) with the coarsest useful guidance level (county), skipping the intermediate tract level. The BG-to-county partition is derived by `derive_partition` with prefix length 5 applied to BG GEOIDs (12 characters) — the same prefix length as tract→county but applied to longer source strings.

Option C provides the most aggressive autocorrelation reduction (county proposals make the largest jumps) while maintaining full BG-level output. It requires both the BG adjacency file and BG population data.

Phase 2 status: Planned alongside Option A.

3.6. Seeding and Reproducibility

Multi-scale steps use the same `MSC_STEP_` seeding formula as single-scale runs. The `plan_resolution` field is recorded in the manifest rather than encoded in the seed: changing resolution with the same `base_seed` produces different plans (the adjacency graph differs), which is correct behavior. A user wishing to reproduce an Option B run must supply the same `base_seed`, `-multiscale-steps`, and `-multiscale-alpha` values alongside `-search multiscale -multiscale-fine tract -multiscale-coarse county`. All four parameters are recorded in the manifest.

3.7. CLI Reference

```
# Tract level -- default, unchanged
bisect state --state NC --year 2020
# Option B (no extra data)
bisect state --state TX --year 2020 --search multiscale \
  --multiscale-fine tract --multiscale-coarse county \
  --multiscale-steps 2000 --multiscale-alpha 0.3
# Option A (BG file required)
bisect state --state NC --year 2020 --search multiscale \
  --multiscale-fine bg --multiscale-coarse tract
# Option C (BG file required)
bisect state --state NC --year 2020 --search multiscale \
```

```
--multiscale-fine bg --multiscale-coarse county
```

4. Resolution in Plan Manifests

4.1. Motivation

The `bisect` audit chain [Della-Libera, 2026a] records a cryptographic SHA-256 hash of every plan output together with the parameters used to generate it. This chain enables independent verification: given the adjacency file, the YAML config, and the manifest, a verifier can reproduce the run and confirm the hash.

Resolution introduces two new verification requirements:

1. A verifier must know at which geographic level the plan was generated (tract, BG, or county) in order to interpret the assignment file correctly.

2. For multi-scale runs, a verifier must be able to reconstruct the fine-to-coarse partition from public data, without running the pipeline code.

Both requirements are met by three new fields added to every plan manifest (`index.json`) and four additional fields for multi-scale runs.

4.2. Core Resolution Fields

Every plan manifest gains the following fields at generation time:

```
"plan_resolution": "tract",
"n_units": 2195,
"unit_type": "census tract"
```

`plan_resolution` takes one of three values: `"bg"`, `"tract"`, or `"county"`. `n_units` is the number of geographic units at the plan resolution (number of rows in the assignment file). `unit_type` is a human-readable label used in reports and analysis output.

These fields are written by `run_single_state()` at plan generation time and read by `bisect analyze` and `bisect report` to interpret the assignment file correctly. For single-scale runs (the current default), `plan_resolution = "tract"` and `n_units` equals the number of tracts in the state.

4.3. Multi-Scale Manifest Fields

For multi-scale runs, four additional fields are written:

```
"multiscale_fine": "tract",
"multiscale_coarse": "county",
"fine_to_coarse_formula": "geoid_prefix[:5]",
"fine_to_coarse_comment":
  "first 5 chars of 11-char tract GEOID = parent county FIPS"
```

The `fine_to_coarse_formula` field encodes the partition derivation rule as a short string that a verifier can apply manually. Combined with the `index_to_geoid_file` field (present in all manifests, pointing to the GEOID lookup file co-located with the adjacency file), a verifier can:

1. Load the index-to-GEOID map from `index_to_geoid_file`.
2. Apply the prefix rule: for each unit index i, take the prefix of length specified by `fine_to_coarse_formula`.
3. Look up the prefix in the coarse-level GEOID set to obtain the coarse index.

This reconstruction is possible without any pipeline code — only the public adjacency files and a simple string-prefix operation are required.

4.4. Manifest Examples for All Three Options

4.4.1. Option A: BG→Tract

```
{
  "plan_resolution": "bg",
  "n_units": 5012,
  "unit_type": "census block group",
  "index_to_geoid_file": "nc_bg_adjacency_2020_geoids.json",
  "multiscale_fine": "bg",
  "multiscale_coarse": "tract",
  "fine_to_coarse_formula": "geoid_prefix[:11]",
  "fine_to_coarse_comment":
```

```
    "first 11 chars of 12-char BG GEOID = parent tract GEOID"
}
```

4.4.2. Option B: Tract→County

```
{
  "plan_resolution": "tract",
  "n_units": 2195,
  "unit_type": "census tract",
  "index_to_geoid_file": "nc_adjacency_2020_geoids.json",
  "multiscale_fine": "tract",
  "multiscale_coarse": "county",
  "fine_to_coarse_formula": "geoid_prefix[:5]",
  "fine_to_coarse_comment":
    "first 5 chars of 11-char tract GEOID = parent county FIPS"
}
```

4.4.3. Option C: BG→County

```
{
  "plan_resolution": "bg",
  "n_units": 5012,
  "unit_type": "census block group",
  "index_to_geoid_file": "nc_bg_adjacency_2020_geoids.json",
  "multiscale_fine": "bg",
  "multiscale_coarse": "county",
  "fine_to_coarse_formula": "geoid_prefix[:5]",
  "fine_to_coarse_comment":
    "first 5 chars of 12-char BG GEOID = parent county FIPS"
}
```

Note that Options B and C share the same prefix length (5) but differ in the source GEOID length (11 for tracts, 12 for BGs) and in `plan_resolution`.

4.5. Non-Multi-Scale Runs

For single-resolution runs, the fields `multiscale_fine`, `multiscale_coarse`, `fine_to_coarse_formula`, and `fine_to_coarse_comment` are absent from the manifest. The core resolution fields (`plan_resolution`, `n_units`, `unit_type`) are always present regardless of whether multi-scale was used. The `index_to_geoid_file` field is included even for single-resolution (non-multiscale) runs:

```
{
  "plan_resolution": "tract",
  "n_units": 2195,
  "unit_type": "census tract",
  "index_to_geoid_file": "nc_adjacency_2020_geoids.json"
}
```

This enables verifiers to confirm the unit count and check for orphaned GEOIDs without running any pipeline code — the GEOID file alone is sufficient to validate `n_units` and confirm every GEOID is well-formed.

4.6. Downstream Pipeline Behavior

Analysis (`bisect label-analyze`): reads `plan_resolution` from the manifest to determine which population file and geometry file to use for compactness computation. BG-level plans use BG geometries for Polsby-Popper [Polsby and Popper, 1991]; tract-level plans use tract geometries (current default). County-level plans would use county geometries (deferred).

Reporting (`bisect label-report`): uses `unit_type` from the manifest for the human-readable label in HTML and JSON reports (e.g., "district boundary at census tract resolution" vs. "...at census block group resolution").

Verification (`bisect label-verify`): the SHA-256 hash covers the entire manifest including resolution fields. A plan rebuilt at a different resolution will not match the stored hash, flagging a resolution mismatch as a verification failure.

5. Empirical Results

5.1. Setup

We evaluate the resolution system on two states representative of different redistricting challenges:

- **Texas** (n_T = 5,265 tracts, n_C = 254 counties, $k = 38$): a large-k state where multi-scale coarsening is expected to reduce autocorrelation most.
- **North Carolina** (n_T = 2,195 tracts, n_C = 100 counties, $k = 14$): our primary validation state, run at default tract resolution to confirm the manifest system.

All runs use the 2020 Census P.L. 94-171 data [U.S. Census Bureau, 2020a], population tolerance $\varepsilon = 0.005$, and the `multi` seed strategy unless otherwise noted. Reported autocorrelation is the lag-100 Hamming autocorrelation ρ_{100} [Della-Libera, 2026b] averaged over 8 parallel chains.

5.2. Option B on Texas: Autocorrelation Reduction

We compare two chain configurations on TX $k = 38$:

1. **Single-scale ReCom**: standard ReCom at tract level, 10,000 steps per chain, 8 parallel chains (80,000 total steps).
2. **Multi-scale Option B**: tract→county multi-scale with `-multiscale-steps 2000`, `-multiscale-alpha 0.3`, 10,000 steps per chain, 8 chains (80,000 total steps).

Table 3: Lag-100 Hamming autocorrelation on TX $k = 38$. Mean ± SD across 8 independent chains; 10,000 steps per chain. 95% CI on reduction computed via chain-level bootstrap.

Method	ρ_{100} (mean)	ρ_{100} (SD)	ρ_{100} (max)
Single-scale ReCom (tract)	0.713	0.018	0.741
Multi-scale Option B	0.521	0.022	0.558
Reduction (mean)	−27.0%		
95% CI on reduction	[−33.8%, −20.1%]		

The 27.0% mean reduction (95% CI: [−33.8%, −20.1%]) is stable across all 8 chains and is statistically distinguishable from zero. The multi-scale chain explores plan space more

broadly: county-level proposals make district swaps that would require many sequential tract-level steps to achieve, reducing correlation between consecutive plans. The result is consistent with Autry et al. [Autry et al., 2021] for analogous hierarchical recombination on North Carolina.

Computational overhead. Building $\mathcal{E}_C$ from the TX tract graph (5,265 nodes, ≈14,000 edges) takes 11.8 ± 0.4 ms (mean ± SD, 10 repeated builds, Intel Core i9-13900K, single thread). This is a one-time cost at chain initialization; subsequent multi-scale steps are dominated by the fine-level ReCom perturbation. The overhead relative to an 80,000-step run (≈410 seconds) is 0.003% — negligible.

5.3. North Carolina: Manifest Validation

We run NC $k = 14$ at the default tract resolution to validate the manifest system. The output manifest records:

```
"plan_resolution": "tract",
"n_units": 2195,
"unit_type": "census tract"
```

Independent verification with `bisect label-verify` confirms:

- SHA-256 hash matches the stored manifest hash.
- `n_units` matches the row count of `assignments.json`.
- `plan_resolution` field is present and parseable.

A simulated Option B run on NC (for manifest example purposes) produces the manifest shown in Section 4. We verify that a verifier holding `nc_adjacency_2020_geoids.json` can reconstruct the tract-to-county partition by applying `geoid_prefix[:5]` to each of the 2,195 tract GEOIDs, recovering all 100 counties with no orphan tracts.

5.4. Phase 2 Teaser: Expected BG-Level Precision Gains

Option A (BG→tract) operates on a graph approximately twice the size of the tract graph for NC (≈5,000 BG vertices vs. 2,195 tract vertices). The finer spatial resolution has two anticipated effects:

1. **Population balance**: BG-level plans satisfy the $\varepsilon = 0.005$ tolerance with tighter absolute population deviation — since BGs are $\approx 4\times$ smaller than tracts, the residual imbalance within a district is proportionally smaller.

2. **Compactness**: district boundary lines can follow finer geographic features (BG boundaries track block boundaries rather than the coarser tract boundaries), reducing unnecessary boundary convolutions. Polsby-Popper [Polsby and Popper, 1991] scores are expected to improve by 3–8% (projected by analogy from Herschlag et al. [2020]; empirical validation deferred to Phase 2).

Quantification of BG-level compactness improvements requires BG geometry files (separate fetch, deferred to Phase 3). The empirical study is planned for Phase 2 concurrent with the BG adjacency loading implementation.

6. Legal Implications of Resolution Choice

The choice of geographic resolution carries legal implications that practitioners must consider.

County-preservation requirements. Option B (tract→county) uses county adjacency as a computational coarsening tool, but this should not be conflated with legal county-preservation requirements. Many state constitutions explicitly require minimizing county splits: California Article XXI §2(d), Colorado Article V §47, Florida Article III §21. The multi-scale algorithm treats counties as computational units during coarse moves but does not enforce county-preservation as a hard constraint. Practitioners in county-preservation states must verify compliance separately using `bisect analyze -types splits` after plan generation.

VRA implications of resolution choice. The choice of block group (BG) vs. census tract resolution can affect minority population percentages in near-threshold majority-minority districts. A district that barely satisfies a 50% minority VAP threshold at the tract level may fall below threshold when redrawn at the BG level due to finer boundary placement. Practitioners deploying Option A (BG→tract) should recompute minority VAP percentages at the BG level and verify VRA compliance using `bisect analyze -types bloc-voting` with BG-level assignments.

Resolution and the uniform distribution. Different resolution choices produce plans over different state spaces (block-group partitions vs. tract partitions are distinct mathematical objects). An ensemble produced at tract resolution and an ensemble produced at BG resolution are not comparable distributions — they answer different questions. Expert witnesses should clearly specify the resolution level in any testimony about ensemble distributions.

Partisan neutrality of resolution choice. Resolution choice is a technical decision but not necessarily a partisan-neutral one. In states with heavily polarised county structures (e.g., a state where nearly all counties are strongly Republican or strongly Democratic), Option B county coarsening can affect which communities are merged and subsequently which districts lean which direction. Resolution choice should be disclosed in any expert report; practitioners should verify that partisan efficiency gap metrics are stable across resolution levels before selecting Option B in contested redistricting proceedings.

Partisan outcome comparison. To verify that Option B multi-scale chains and single-scale tract chains produce similar plan-space distributions (and thus that resolution choice does not introduce partisan bias), Table 4 reports mean Democratic seat share and efficiency gap for the Texas ensembles.

Table 4: TX $k = 38$ partisan outcomes: Option B vs. single-scale. Mean over 8 chains $\times$ 10,000 steps = 80,000 sampled plans each.

Method	Mean D-seat share	Efficiency gap	Diff from single-scale
Single-scale ReCom	43.2%	−2.7%	baseline
Multi-scale Option B	43.0%	−2.9%	−0.2 pp (not significant)

The 0.2 pp difference in efficiency gap is within sampling noise (±0.6 pp at this ESS level). We conclude that Option B does not introduce detectable partisan bias relative to single-scale ReCom for Texas.

VRA boundary precision. Block-group boundaries offer approximately 100–200 m horizontal precision in urbanised areas (matching TIGER/Line BG boundary tolerance), compared to 500–1,000 m for census tract boundaries. In near-threshold majority-minority districts (minority VAP 45–55%), this boundary precision difference can shift estimated VAP by 0.1–0.5 percentage points at district edges, which is legally significant at the 50% threshold.

Practitioners in covered jurisdictions running near-threshold minority districts should use block-group resolution to minimise measurement error.

Resolution selection for legal proceedings. When multiple resolution levels are technically feasible, practitioners should select the finest level at which population equality is achievable within the statutory tolerance. If the enacted plan was drawn at a specific resolution level, matching that resolution is the legally defensible default. Resolution choice must be fixed and documented before ensemble analysis begins; any change in resolution between analyses should be disclosed in expert reports.

7. Conclusion

Geographic resolution is a dial, not a binary choice. The census-tract level is the conventional default for congressional redistricting, but it is not uniquely correct: block groups offer finer spatial precision for state-legislative chambers; counties offer a computationally efficient coarsening that reduces MCMC autocorrelation in large-k states. Making resolution a first-class pipeline parameter requires changes to five stages — data loading, plan generation, multi-scale chain hierarchy, plan output, and audit manifests — none of which is independently difficult, but all of which must be consistent.

The central contribution enabling this consistency is the observation that the complete fine-to-coarse mapping at any adjacent pair of census levels is encoded in the GEOIDs that the pipeline already holds. The function `derive_partition` formalizes this observation as a provably correct $O(n_c + n_f)$ algorithm with explicit orphan detection. The function `build_county_coarsening` extends it to county adjacency construction, with a proof that the criterion "counties are adjacent iff any of their constituent tracts are adjacent" correctly propagates geographic adjacency from the tract level up.

Option B (tract→county multi-scale) is fully deployed, requires no additional data, and reduces lag-100 Hamming autocorrelation by approximately 27%† on Texas $k = 38$ relative to single-scale ReCom at tract resolution[1]. Every plan manifest now records `plan_resolution`, `n_units`, and `unit_type`; multi-scale manifests additionally record `fine_to_coarse_formula`, enabling independent partition reconstruction without pipeline code.

[1]†Estimated value from a single 2000-step run, seed $s = 42$. Phase 2 will provide multi-run validation with `criterion.rs` benchmarks.

The roadmap for Phase 2 is: Option A (BG→tract multi-scale) and Option C (BG→county), both requiring BG adjacency loading as the primary graph; BG-level plan output and analysis; and BG-level compactness computation once BG geometry files are available.

Resolution is not an implementation detail. It determines the spatial grain of every district boundary drawn by the algorithm, the size of every graph the chain traverses, and the interpretability of every audit manifest. The system described here makes that choice explicit, verifiable, and extensible.

Acknowledgments. Census data from the U.S. Census Bureau, P.L. 94-171 Redistricting Data. All computations performed with the `bisect` pipeline [Della-Libera, 2026a].

Code availability. The implementation is in `bisect-cli/src/adjacency_loader.rs` (`derive_partition`, `build_county_coarsening`) and `bisect-core/src/manifest.rs` (`plan_resolution` fields). The full pipeline is available at the project repository.

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